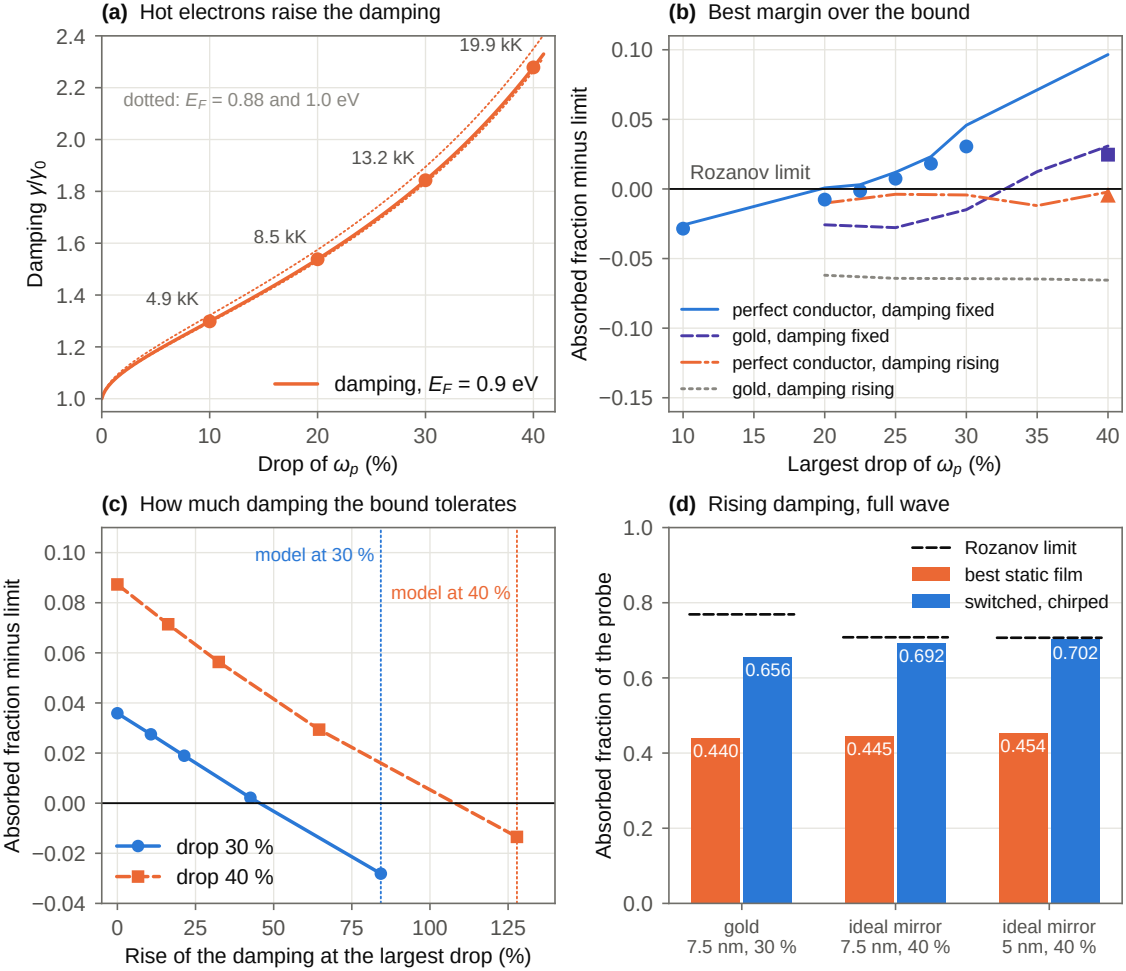
(a) Hot electrons raise the damping
Damping $\gamma/\gamma_0$
Drop of $\omega_p$ (%)
dotted: $E_F$ = 0.88 and 1.0 eV
4.9 kK
8.5 kK
13.2 kK
19.9 kK
damping, $E_F$ = 0.9 eV
(b) Best margin over the bound
Absorbed fraction minus limit
Largest drop of $\omega_p$ (%)
Rozanov limit
perfect conductor, damping fixed
gold, damping fixed
perfect conductor, damping rising
gold, damping rising
(c) How much damping the bound tolerates
Absorbed fraction minus limit
Rise of the damping at the largest drop (%)
model at 30 %
model at 40 %
drop 30 %
drop 40 %
(d) Rising damping, full wave
Absorbed fraction of the probe
Rozanov limit
best static film
switched, chirped
0.440
0.656
0.445
0.692
0.454
0.702
gold
7.5 nm, 30 %
ideal mirror
7.5 nm, 40 %
ideal mirror
5 nm, 40 %